MATHEMATICS

SECTION – I
Single Correct Choice Type

This section contains 6 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

---

1. From a point P outside a circle with centre at C, tangents PA and PB are drawn such that $\frac{1}{(CA)^2}+\frac{1}{(PA)^2}=\frac{1}{16}$, then the length of chord AB is

(a) 8 (b) 12 (c) 16 (d) none of these

2. The coefficient of x^2 in $(1+x)(1+2x)(1+4x).....(1+2^n x)$ is

(a) $\frac{(2^n+1)(2^n+2)}{3}$ (b) $\frac{(2^{n+1}-1)(2^{n+1}-2)}{3}$

(c) $\frac{(2^{n+1}+1)(2^{n+1}+2)}{3}$ (d) $\frac{(2^{n+1}+1)(2^{n+1}-2)}{5}$

3. If the function $f(x)=\left(4\sin^2 x-1\right)^n\left(x^2-x+1\right), n\in N$ has a local minimum at $x=\frac{\pi}{6}$, then

(a) n can be any odd number
(b) n can only be an odd prime number
(c) n can be any even number
(d) n must be a multiple of four

4. Let $I_1=\int_0^1 \frac{e^x}{1+x}dx$ and $I_2=\int_0^1 \frac{x^2dx}{e^{x^3}(2-x^3)}$. Then $\frac{I_1}{I_2}$ is

(a) $\frac{3}{e}$ (b) $\frac{e}{3}$ (c) $3e$ (d) $\frac{1}{3e}$

5. Suppose $f(x) = x^3 + ax^2 + bx + c$, where *a*, *b*, *c* are chosen respectively by throwing a die three times. Then the probability that $f(x)$ is an increasing function, is

(a) $\frac{4}{9}$ (b) $\frac{3}{8}$ (c) $\frac{2}{5}$ (d) $\frac{16}{34}$

6. ΔABC is inscribed in a unit circle. The three bisectors of angles *A*, *B* and *C* are extended to intersect the circle at A_1, B_1 and C_1 respectively.

Then the value of $\frac{AA_1 \cos\frac{A}{2} + BB_1 \cos\frac{B}{2} + CC_1 \cos\frac{C}{2}}{\sin A + \sin B + \sin C}$ is

(a) 2 (b) 4

(c) 6 (d) 8

SECTION – II
Multiple Correct Choice Type

This section contains 5 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

---

7. The values of *a*, for which $\frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8} + \frac{a}{30} = 0$ doesn't have a real solution, are

(a) – 10 (b) 12 (c) 5 (d) – 30

8. If $f(x + y) = f(x) f(y)$ for all $x, y \in R$, $f(1) = 2$ and $a_r = f(r)$ for $r \in N$, then the co-ordinates of a point on the parabola $y^2 = 8x$ whose focal distance is 4, may be

(a) (a_1, a_2) (b) $(a_1, -a_2)$

(c) $(a_3, -a_3)$ (d) (a_3, a_3)

9. If $\int \frac{dx}{1-\sin^4 x} = \frac{1}{2}\tan x + A\tan^{-1}(f(x)) + C$, then

(a) $A = \sqrt{2}$ (b) $A = \frac{1}{2\sqrt{2}}$

(c) $f(x) = \frac{\tan x}{\sqrt{2}}$ (d) $f(x) = \sqrt{2}\tan x$

10. If the unit vectors $\vec{a}$ and $\vec{b}$ are inclined at an angle 2θ such that $|\vec{a} - \vec{b}| < 1$ and $0 \le \theta \le \pi$, then θ lies in the interval

(a) $\left[0, \frac{\pi}{6}\right)$ (b) $\left(\frac{5\pi}{6}, \pi\right]$

(c) $\left[\frac{\pi}{6}, \frac{\pi}{2}\right]$ (d) $\left[\frac{\pi}{2}, \frac{5\pi}{6}\right]$

11. Let $\alpha = \cos\frac{2\pi}{11} + i\sin\frac{2\pi}{11}$, $\lambda = \alpha^6$, $\mu = \alpha^7$, $\beta = \alpha^2$, then

(a) $\text{Re}(\lambda + \lambda^2 + \lambda^3 + \lambda^4 + \lambda^5) = -\frac{1}{2}$

(b) $(\mu - \beta)(\mu - \beta^2)(\mu - \beta^3)\ldots(\mu - \beta^9)(\mu - \beta^{10}) = 0$

(c) $(i - \beta)(i - \beta^2)(i - \beta^3)\ldots(i - \beta^{10}) = i$

(d) $(\alpha - \beta)(\alpha - \beta^2)\ldots\ldots(\alpha - \beta^9)(\alpha - \beta^{10}) = 1$

SECTION – III
Reasoning Type

This section contains 5 reasoning type questions numbered 12 to 16. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

12. **Statement-1** : $f(x)=[x]^2+\sqrt{\{x\}^2}$ is discontinuous at all integers (where [.] denotes greatest integer function and {.} denotes fractional part function).

Statement-2 : $f(x)=[x]$ is discontinuous at all integers.

(a) A (b) B (c) C (d) D

13. **Statement-1** : If positive numbers x, y, z satisfy $x + y + z = 1$, then $\left(\frac{(1-2x)+(1-2y)+(1-2z)}{3}\right) \geq ((1-2x)(1-2y)(1-2z))^{\frac{1}{3}}$

Statement-2 : For any 3 positive numbers a, b, c their A.M. ≥ G.M.

(a) A (b) B (c) C (d) D

14. **Statement-1** : Any positive integral power of $(\sqrt{2}-1)$ can be expressed as $\sqrt{N}-\sqrt{N-1}$ for some natural number $N > 1$.

Statement-2 : Any positive integral power of $(\sqrt{2}-1)$ can be expressed as $A+B\sqrt{2}$ where A and B are integers.

(a) A (b) B (c) C (d) D

15. **Statement-1** : Let $A = \{x \mid x$ is a prime number and $x < 30\}$. Then the number of different rational numbers, whose numerator and denominator belong to A is 93.

Statement-2 : $\frac{p}{q}$ is a rational number $\forall q \neq 0$ and $p, q \in I$.

(a) A (b) B (c) C (d) D

16. **Statement-1** : The number of solution(s) of the equation

$$(\sin x - 2\cos x) = \min\left\{\sqrt{3},\ a^2 - 4a + 5\right\} \text{ is 2.}$$

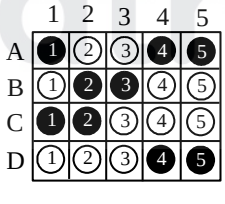
Statement-2 : Range of $\min\left(\sqrt{3},\ a^2 - 4a + 5\right)$ is $[1,\ \sqrt{3}]$ for $a \in R$.

(a) A (b) B (c) C (d) D

SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1. $f(x)=\dfrac{2^x+2^{-x}}{2^x-2^{-x}}$

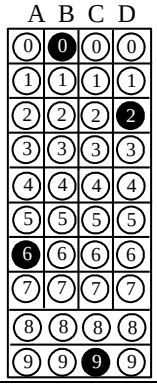
Column- I	Column -II
(A) If $1<x<2$, then $f(x)$ satisfies	**1.** $f(x)>1$
(B) If $0<x<\dfrac{1}{2}$, then $f(x)$ satisfies	**2.** $f(x)<-1$
(C) If $-\dfrac{1}{2}<x<0$, then $f(x)$ satisfies	**3.** $f(x)>3$
(D) If $-2<x<-1$, then $f(x)$ satisfies	**4.** $f(x)<-3$ **5.** $-1<f(x)<1$

2. A matrix of order 3 × 3 is formed by using the elements {1, 2, ……9} without repetition. Considering this; match column I and II (here $a = 2$, $b = 3$, $c = 5$)

Column- I	Column -II
(A) Probability that first row of the matrix has entries in increasing order, is	**1.** $\dfrac{1}{ab}$
(B) Probability that first row and first column have entries in increasing order, is	**2.** $\dfrac{1}{a^2c}$
(C) Probability that first row and first column have entries in increasing order given that last entry of the third row and third column is the greatest, is	**3.** $\dfrac{a^2+c}{a^2b^2c}$
(D) Probability that first row of matrix has entries in increasing order given that entries of second row and third row are in increasing order, is	**4.** $\dfrac{c-a}{ab^2}$ **5.** $\dfrac{1}{a}-\dfrac{1}{b}$

SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



1. Two lines $zi - \bar{z}\,i + 2 = 0$ and $z(1+i) + \bar{z}(1-i) + 2 = 0$ intersect at a point P. There is a complex number $\alpha = x + iy$ at a distance of 2 units from the point P which lies on line $z(1+i) + \bar{z}(1-i) + 2 = 0$. Find $[|x|]$ (where $[\cdot]$ represents greatest integer function).

Space for work

mathongo

2. Let P, Q be two points on the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ whose eccentric angles differ by a right angle. Tangents are drawn at P and Q to meet at R. If the chord PQ divides the join of C and R in the ratio $m : n$ (C being centre of ellipse), then find $m + n$ ($m : n$ is in simplified form).

Space for work

mathongo

3. Find the value of $\frac{29\int_0^1 \left(1-x^4\right)^7 dx}{4\int_0^1 \left(1-x^4\right)^6 dx}$.

Space for work

mathongo

4. If $f(x)=a\cos(\pi x)+b,\ f'\left(\frac{1}{2}\right)=\pi$ and $\int_{1/2}^{3/2} f(x)dx=\frac{2}{\pi}+1$, then find the value of $-\frac{12}{\pi}\left(\sin^{-1}a+\cos^{-1}b\right)$.

Space for work

mathongo